

Question	Answer	Marks
5(a)(i)	Square loop rotate about axis CD.	A1
5(a)(ii)	Square loop will rotate clockwise when viewed from C. The current in sides ab and bc of square loop interact with the components of the magnetic flux density that is perpendicular to the respective sides and hence experience a force. By Fleming's Left Hand Rule, the forces acting on sides ab and bc acts <i>in the direction out of the paper</i>. Similarly, the forces acting on sides cd and da can be determined to act <i>in the direction into the paper</i>. Thus, <i>there is a couple</i> resulting in clockwise torque (viewed from C) about the axis CD.	A1 B1 B1 B1
5(b)(i)	When a and c are 2.00 m apart, the area enclosed by the loop consists of four triangular sections, each having hypotenuse of 3.00 m, height of 1.00 m, and base of $\sqrt{3.00^2 - 1.00^2} = 2.83$ m The decrease in the enclosed area is $\Delta A = A_i - A_f = (3.00)^2 - 4[\frac{1}{2}(1.00)(2.83)] = 3.34 \text{ m}^2$ The average induced e.m.f. is $\varepsilon = \frac{\Delta \phi}{\Delta t} = B \frac{\Delta A}{\Delta t} = \frac{(0.100)(3.34)}{0.100} = 3.34 \text{ V}$	B1 B1 B1 A1
5(b)(ii)	The induced current in the loop, $I = \frac{\varepsilon}{R} = \frac{3.34}{10.0} = 0.334 \text{ A}$	A1
	Max Marks	10

Setter: Jit Ning Marker:		
Question	Answer	Marks
6(a)	Frequency = 50 Hz	A1